Hakan Alex Kucukhuseyin

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EDUCATION

Drexel University Philadelphia, PA
B.S. Electrical Engineering & Computer Engineering (Double Major), Minor in Finance *Expected June 2028*
GPA: 3.6 | **Finance Minor GPA:** 4.0
Honors: Dean's List (September 2024 – Present) | 1st Place, Drexel Freshman Engineering Design (2025)
Relevant Coursework: Data Structures, Circuits II, Signals & Systems, Microcontrollers, Probability & Data Analytics

EXPERIENCE

Blues Yachting Mugla, Turkey
Software Developer *June 2023 – June 2024 | June 2025 – August 2025*
- Built a React/Node.js/PostgreSQL operations platform that replaced spreadsheet workflows across quoting, CRM, inventory, and payments with 18 integrated modules.
- Cut quote preparation from 15–20 minutes to a few minutes (∼85%) through automated Turkish tax calculations, multi-line pricing, and one-click PDF generation.
- Deployed with Docker and JWT authentication; set up GitHub Actions CI that runs lint, unit, and end-to-end tests against a disposable PostgreSQL database on every push.
- Partnered with sales, service, and operations staff to gather requirements, prioritize workflows, train users, and iterate on product feedback.

PROJECTS

VitalView – Non-Invasive Glucose Monitor | Demo: hakanalex-k.github.io/vitalview Philadelphia, PA
Team Lead, Freshman Engineering Design (1st Place) *March 2025 – June 2025*
- Led an end-to-end prototype combining ESP32/PPG sensing, a Node.js API, TensorFlow/Keras inference, and a React dashboard for glucose and heart-rate monitoring.
- Trained a model on 600-feature IR/red signal vectors; added checks to reject low-amplitude or flatline readings.

UniTR – Turkish University Explorer with AI Search Philadelphia, PA
Sole Developer, Personal Project *January 2025 – Present*
- Built a Python ETL pipeline ingesting 12,239 programs across 224 universities from the YÖK Atlas API (2023–2026 admissions) into a validated, checksummed dataset, with optional PostgreSQL import.
- Built a Next.js/TypeScript site and Express API for rank-based search, program history charts, and university comparison; one query engine serves in-memory and PostgreSQL data with identical results, backed by 60+ tests.
- Developed a streaming LLM search assistant on the Claude API with tool calling and server-verified citations.
- Built a 20-case LLM evaluation suite covering tool accuracy, grounding, and prompt injection; raised Claude Haiku's pass rate from 14/20 to 20/20, matching Opus on the suite at ∼30× lower estimated token cost per query.

Hack Computer – CPU & Memory System (Verilog) Philadelphia, PA
Course Project, ECE 350 *June 2026 – August 2026*
- Built a stored-program Hack CPU in Verilog: ALU, registers, PC, control logic, RAM/ROM, and memory-mapped I/O.

Drexel Turkish Student Association Website | Live site: drexeltsa.vercel.app *October 2026*
- Built and deployed a full-stack Next.js/React/TypeScript site (Vercel, Neon Postgres, $0/month) with an admin dashboard where board members publish events, blog posts, photo galleries, and team profiles.
- Secured the site with scrypt password hashing, hashed server-side sessions, CSRF origin checks, a nonce-based Content Security Policy, and database-backed rate limiting.

TECHNICAL SKILLS

Languages	C, Python, JavaScript/TypeScript, SQL, Verilog
Software	React, Node.js, Express, TensorFlow/Keras, PostgreSQL, Docker, Git, GitHub Actions, Linux, REST APIs
AI	LLM APIs (Claude, OpenAI), tool calling, streaming, prompt caching, LLM evals
Hardware	ESP32, microcontrollers, digital logic, I2C, computer architecture

ACTIVITIES

IEEE, Drexel Chapter – Member September 2024 – Present
Drexel Turkish Student Association – Event Coordinator